



DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

June 28, 2016

Name, Surname:

Student ID:

Row:

Column:

Time: 2 hour 15 minutes

Prof. Marcello Restelli

Maximum Mark: 31

- The following exam is composed of **10 exercises** (one per page). The first page need to be filled with your **name, surname and student ID**. The following pages should be used only in the large squares present on each page. Any solution provided outside these spaces will not be considered for the final mark.
- During this exam you are **not allowed** to use electronic devices like laptops, smartphones, tablets and similar. As well you are not allowed to bring with you any kind of note, book, or written scheme. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above mentioned rules will imply the immediate expulsion of the student from the exam room and the **direct annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Ex. 9	Ex. 10	Tot.
/ 5	/ 5	/ 5	/ 2	/ 2	/ 2	/ 2	/ 2	/ 3	/ 3	/ 31

Exercise 1 (5 points)

Describe the supervised learning technique denominated Support Vector Machines.

Exercise 2 (5 points)

Describe the unsupervised learning technique denominated K-means.

Exercise 3 (5 points)

Describe the differences existing between the Q-learning and SARSA algorithms.

Exercise 4 (2 point)

Considering the problem of modeling the poker game, tell if it has:

1. Finite/infinite actions;
2. Deterministic/stochastic transitions;
3. Deterministic/stochastic rewards;
4. Finite/indefinite/infinite horizon;
5. Stationary/non-stationary environment.

Motivate your answers.

Exercise 5 (2 point)

Why is linear regression important to understand? Select all that apply and justify your choice:

1. The linear model is often correct;
2. Linear regression is extensible and can be used to capture nonlinear effects;
3. Simple methods can outperform more complex ones if the data are noisy;
4. Understanding simpler methods sheds light on more complex ones;
5. A fast way of solving them is available.

Exercise 6 (2 point)

While you fit a Linear Model to your data set. You are thinking about changing the Linear Model to a Quadratic one (i.e., a Linear Model with quadratic features $\phi(x) = [1, x, x^2]$). Which of the following is most likely true:

1. Using the Quadratic Model will decrease your Irreducible Error;
2. Using the Quadratic Model will decrease the Bias of your model;
3. Using the Quadratic Model will decrease the Variance of your model;
4. Using the Quadratic Model will decrease your Reducible Error.

Provide motivations to your answers.

Exercise 7 (2 point)

Consider $x, y \in \mathbb{R}^d$, which ones of these are similarity measure:

1. $k(x, y) = x^T y$ (dot product);
2. $k(x, y) = x^T y + (x^T y)^2$;
3. $k(x, y) = k_1(x, y) + ck_2(x, y) \times k_3(x, y)$, where k_1, k_2 and k_3 are valid kernels in $\mathbb{R}^d$;
4. $k(x, y) = \log(x)e^{-3y}$ ($d = 1$);
5. $k(x, y) = x^T Ay$ with $A = \begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix}$ ($d = 2$);
6. $k(x, y) = \sqrt{(1 - \cos^2(x))} \cos(y - \pi/2)$, ($d = 1$).

Exercise 8 (2 point)

Tell if the following statements are true or false and motivate your answers.

1. With MC estimation you can extract a number of samples for the value function equal to the length of the episode you consider for prediction;
2. Generally, every-visit estimation is better if you use a small amount of episodes;
3. Stochasticity in the rewards requires the use of a larger number of episodes to have precise prediction of the MDP value by resorting to MC estimation ;
4. MC estimation works better than TD if the problem is not Markovian.

Exercise 9 (3 points)

Suppose you have trained a logistic regression classifier whose output is y_n . You predict 1 if $y_n > \tau$, and predict 0 if $y_n < \tau$, where currently the threshold is $\tau = 0.5$.

Suppose you decrease the threshold to $\tau = 0.3$. Which of the following statements are true? Check all that apply and provide a motivation.

1. The classifier is likely to now have higher precision.
2. The classifier is likely to have unchanged precision and recall, but higher accuracy.
3. The classifier is likely to now have higher recall.
4. The classifier is likely to have unchanged precision and recall, but lower accuracy.

Exercise 10 (3 points)

Write the formula for the minimum regret we might have on average over $T = \lceil e^{10} \rceil$ time steps in the case we have a stochastic MAB with 3 arms with:

$$R(a_1) = 0.2$$

$$R(a_2) = 0.4$$

$$R(a_3) = 0.7$$

and each distribution $\mathcal{R}(a_i)$ is a Bernoulli.

Note that the KL divergence for Bernoulli variables with means p and q is:

$$KL(p, q) = p \log \left(\frac{p}{q} \right) + (1 - p) \log \left(\frac{(1 - p)}{(1 - q)} \right)$$

Hints: $\log(\frac{0.2}{0.7}) = -1.25$, $\log(\frac{0.8}{0.3}) = 0.98$, $\log(\frac{0.4}{0.7}) = -0.56$ and $\log(\frac{0.6}{0.3}) = 0.69$

Is it possible that your algorithm achieves lower regret? If so, provide an example.